

A Norming Finite-Quotient Criterion for C^* -Algebras with UBCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The scalar-unitization packet handles one multiplier direction over a continuous-trace ideal. This packet gives the finite-dimensional quotient version under the natural hypothesis that the continuous-trace ideal still uniformly norms the extension.

Definition 1. *Let I be a closed ideal of a C^* -algebra A . We say that I is γ -norming in A , where $0 < \gamma \leq 1$, if*

$$\sup\{\|\pi(a)\| : \pi \in \widehat{I}\} \geq \gamma\|a\| \quad (a \in A).$$

Equivalently, the primitive quotients coming from $\text{Prim}(I)$ see at least a fixed fraction γ of every norm in A .

Theorem 1 (norming finite-dimensional quotient criterion). *Let*

$$0 \longrightarrow I \longrightarrow A \xrightarrow{q} F \longrightarrow 0$$

be an exact sequence of C^ -algebras. Assume:*

1. F is finite dimensional;
2. I is continuous trace, and every elementary fibre of I is $\mathcal{K}(H)$ for a separable nonzero Hilbert space H ;
3. $\text{Prim}(I)$ has a countable π -basis;
4. I is γ -norming in A for some $\gamma > 0$.

Then A has UBCP.

Consequently, within this norming finite-quotient class,

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(I) \text{ has a countable } \pi\text{-basis}.$$

Idea of the Proof

Choose a bounded linear section $\sigma : F \rightarrow A$. Every $a \in A$ can be written

$$a = \sigma(q(a)) + k, \quad k \in I.$$

The old compact-bump proof controls the compact section k , but the lifted finite-dimensional quotient part $\sigma(q(a))$ acts locally as an arbitrary bounded operator $B(P)$, not as a compact operator.

The matrix-block replacement for the scalar spike is simple. At a point where $\|\sigma(q(a))(P) + k(P)\|$ is bounded below, approximate $k(P)$ by a finite-rank operator T . Then choose a rank-one operator R aligned with a vector where $B(P) + T$ almost attains its norm. For a large fixed M ,

$$\|B(P) + T + MR\| \approx M + \|B(P) + T\|,$$

whereas along the bump path

$$\|k(P) - s(T + MR)\| \lesssim M \quad (0 \leq s \leq 1).$$

Thus the finite-dimensional quotient part supplies the same extra margin that the scalar phase supplied in the unitization proof.

Proof

We first record the two elementary lemmas used in the covering construction.

Lemma 1 (rank-one ballast). *Let H be a separable nonzero Hilbert space, let $D \in \mathcal{B}(H)$, let $M > 0$, and let $\varepsilon > 0$. There is a norm-one rank-one operator R belonging to a fixed countable norm-dense family of rank-one operators such that*

$$\|D + MR\| \geq M + \|D\| - \varepsilon.$$

Proof. Choose a unit vector ξ with $\|D\xi\| > \|D\| - \varepsilon/3$. If $D\xi \neq 0$, put $v = D\xi/\|D\xi\|$; otherwise take any unit vector v . The rank-one operator $R_0\eta = \langle \eta, \xi \rangle v$ satisfies

$$\|(D + MR_0)\xi\| \geq M + \|D\| - \varepsilon/3.$$

Rank-one norm-one operators are norm separable when H is separable: take initial and final unit vectors from a fixed countable dense subset of the unit sphere. Choose R in that countable family with $\|R - R_0\| < \varepsilon/(3M)$. Then

$$\|D + MR\| \geq M + \|D\| - \varepsilon.$$

□

Lemma 2 (charted bump families). *If I is continuous trace with separable fibres and $\text{Prim}(I)$ has a countable π -basis, then there are countably many charts*

$$\Phi_n : I_{O_n} \longrightarrow C_0(O_n, \mathcal{K}(H_n))$$

with separable nonzero H_n , such that (O_n) is a π -basis of $\text{Prim}(I)$. For each n , there is a countable family $\mathcal{B}_n \subset C_c(O_n)$, $0 \leq h \leq 1$, $\|h\| = 1$, whose supports are cofinal in the nonempty open subsets of O_n .

Proof. This is the standard local triviality and bump selection for continuous-trace algebras, as used in the earlier continuous-trace packet. The spaces O_n are locally compact Hausdorff, so local regularity gives the scalar bumps. $\square$

Proof of Theorem 1. The implication $\text{UBCP} \Rightarrow \text{BCP}$ is immediate. Conversely, if A has BCP, the primitive π -base obstruction gives a countable π -basis for $\text{Prim}(A)$, hence also for the open subspace $\text{Prim}(I)$. Since F is finite dimensional, the finite quotient part of $\text{Prim}(A)$ does not affect countability of a π -basis. It remains to prove UBCP from the four stated assumptions.

Choose a bounded linear section $\sigma : F \rightarrow A$, and put

$$C = \max\{1, \|\sigma\|\}.$$

Let

$$M = C + 2.$$

Choose $0 < \varepsilon < 1$ so small that

$$(2C + 5)\varepsilon < \gamma/2.$$

Because F is finite dimensional, the unit ball of F has a countable ε -net Z .

Fix the chart and bump families from Lemma 2. In each chart choose:

- a countable norm-dense set $\mathcal{T}_n$ of finite-rank operators in $\mathcal{K}(H_n)$;
- a countable norm-dense set $\mathcal{R}_n$ of norm-one rank-one operators on H_n .

For every

$$z \in Z, \quad n, \quad h \in \mathcal{B}_n, \quad T \in \mathcal{T}_n, \quad R \in \mathcal{R}_n$$

form

$$c(z, n, h, T, R) = \sigma(z) + \Phi_n^{-1}(h(T + MR)) \in A.$$

Retain only those countably many centers satisfying

$$\|c(z, n, h, T, R)\| \geq M + \gamma/2 - (C + 3)\varepsilon.$$

For each retained center use the common radius

$$r = M + (C + 2)\varepsilon.$$

The choice of ε gives

$$M + \gamma/2 - (C + 3)\varepsilon - r = \gamma/2 - (2C + 5)\varepsilon > 0,$$

so the retained balls are uniformly off the origin.

We claim that these balls cover S_A . Let $a \in S_A$. Put

$$y = q(a) \in F, \quad k = a - \sigma(y) \in I.$$

Then $\|y\| \leq 1$, so choose $z \in Z$ with $\|y - z\| < \varepsilon$, and set

$$b = \sigma(z), \quad e = \sigma(y - z).$$

Thus $\|e\| \leq C\varepsilon$ and

$$a = b + k + e.$$

By the γ -norming hypothesis, the open set

$$W = \{P \in \text{Prim}(I) : \|a(P)\| > \gamma/2\}$$

is nonempty. Choose a chart $O_n \subset W$, and choose $P_0 \in O_n$. Write $k_n(P) = \Phi_n(k|_{I_{O_n}})(P)$. Choose

$$T \in \mathcal{T}_n \quad \text{with} \quad \|T - k_n(P_0)\| < \varepsilon.$$

By norm-continuity of k_n , choose a nonempty open neighbourhood

$$N \subset O_n \cap W$$

of P_0 such that

$$\|k_n(P) - k_n(P_0)\| < \varepsilon \quad (P \in N).$$

Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$. Since h has compact support and norm one, there is $P_1 \in \text{supp } h$ such that $h(P_1) = 1$.

Let $B(P_1) \in \mathcal{B}(H_n)$ denote the local operator by which b acts in the irreducible representation over P_1 . Since

$$a(P_1) = B(P_1) + k_n(P_1) + e(P_1)$$

and $P_1 \in W$, we have

$$\|B(P_1) + k_n(P_1)\| > \gamma/2 - C\varepsilon.$$

Also $\|k_n(P_1) - T\| < 2\varepsilon$, whence

$$D := B(P_1) + T \quad \text{satisfies} \quad \|D\| > \gamma/2 - (C + 2)\varepsilon.$$

By Lemma 1, choose $R \in \mathcal{R}_n$ such that

$$\|D + MR\| \geq M + \|D\| - \varepsilon > M + \gamma/2 - (C + 3)\varepsilon.$$

For the center $c = c(z, n, h, T, R)$, the local value at P_1 is

$$B(P_1) + T + MR.$$

Therefore c is one of the retained centers.

It remains to show that $a \in B(c, r)$. Let $P \in \text{supp } h$, and put $s = h(P)$. In the chart,

$$a(P) - c(P) = e(P) + k_n(P) - s(T + MR).$$

Since $P \in N$,

$$\|k_n(P) - T\| < 2\varepsilon.$$

Also

$$\|T\| \leq \|k_n(P_0)\| + \varepsilon \leq \|a(P_0)\| + \|\sigma(y)(P_0)\| + \varepsilon \leq 1 + C + \varepsilon < M.$$

Thus

$$\begin{aligned} \|a(P) - c(P)\| &\leq C\varepsilon + 2\varepsilon + \|(1-s)T - sMR\| \\ &\leq C\varepsilon + 2\varepsilon + (1-s)\|T\| + sM \\ &< M + (C+2)\varepsilon = r. \end{aligned}$$

If $P \in \text{Prim}(I) \setminus \text{supp } h$, then the compact bump vanishes at P , and

$$\|a(P) - c(P)\| = \|e(P) + k(P)\| \leq C\varepsilon + \|a(P)\| + \|\sigma(y)(P)\| \leq C\varepsilon + 1 + C < r,$$

because $M = C + 2$ and $\varepsilon < 1$. Finally, on primitive quotients coming from $F = A/I$, the ideal bump vanishes and

$$\|q(a - c)\| = \|y - z\| < \varepsilon < r.$$

Taking the supremum over primitive quotients gives $\|a - c\| < r$. Hence the retained countable family of balls covers S_A , and the uniform gap proved above gives UBCP. $\square$

Scope and Limitations

This is a finite-dimensional quotient theorem, but not yet the arbitrary finite-dimensional quotient theorem. The new assumption is that the continuous-trace ideal is uniformly norming in the extension. That hypothesis is exactly what lets the matrix-ballast argument find a local ideal point where the element has definite size.

The remaining case is the non-norming quotient-dominant case: elements whose norm is seen almost entirely by the finite quotient primitive points while the continuous-trace ideal sees only a small part. The next plausible route is to show that this quotient-dominant part splits off as a finite-dimensional direct summand, or to build a separate quotient-point covering mechanism.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, finite-dimensional quotients, finite-dimensional extensions, unitization, matrix-block ballast, continuous trace, BCP, UBCP, and primitive π -bases. Web searches on June 17, 2026 for BCP finite-codimensional permanence, finite-dimensional C^* -quotient splitting, and Fell/type-I local continuous-trace structure found the source paper, vector-valued $C_0(K, X)$ BCP stability arXiv:2111.04921, Calkin/operator-space BCP papers arXiv:2412.07137 and arXiv:2507.02261, and Fell-algebra structure arXiv:1207.2540, but no exact norming finite-quotient BCP theorem.

Human Review Focus

The main point to check is Lemma 1 and its use at the point P_1 where the bump equals one. Notice that no norm-continuity of the lifted finite-dimensional quotient field is used: the lifted operator $B(P)$ cancels in the distance estimate, and it is used only at the single point P_1 to lower-bound the center norm.

Reviewers should also check the constants involving $C = \max(1, \|\sigma\|)$, γ , $M = C + 2$, and ε , and the outside-support estimate.

References

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- [4] J. Dixmier, *C^* -Algebras*, North-Holland, 1977.
- [5] I. Raeburn and D. P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.